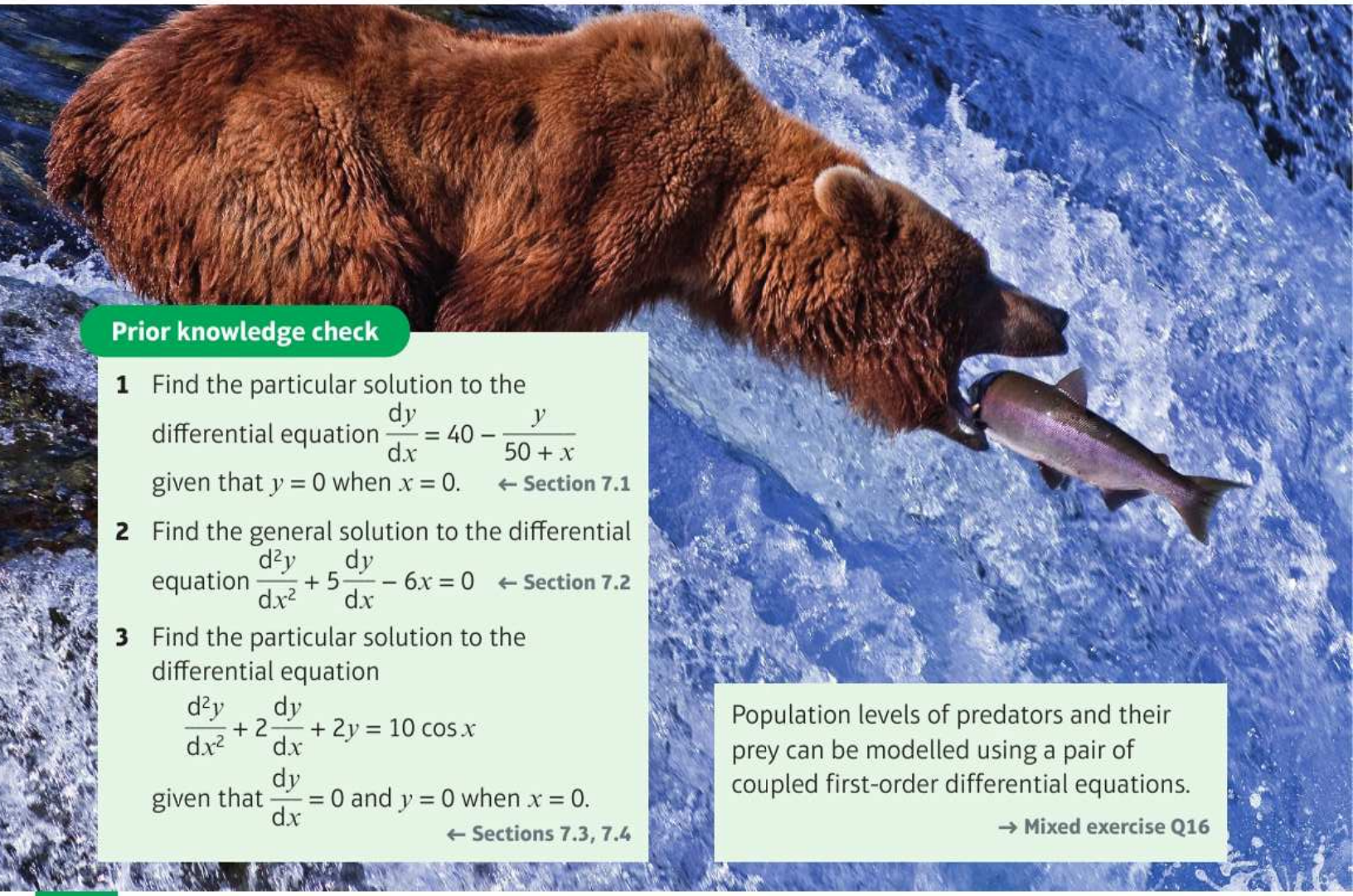
# 8 Modelling with differential equations

## Objectives

After completing this chapter you should be able to:

- Model real-life situations with first-order differential equations → pages 171–175
- Use differential equations to model simple harmonic motion → pages 175–180
- Model damped and forced oscillations using differential equations → pages 180–186
- Model real-life situations using coupled first-order differential equations → pages 186–190



## Prior knowledge check

**1** Find the particular solution to the differential equation $\frac{dy}{dx} = 40 - \frac{y}{50 + x}$ given that $y = 0$ when $x = 0$. ← Section 7.1

**2** Find the general solution to the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} - 6x = 0$ ← Section 7.2

**3** Find the particular solution to the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 10\cos x$$

given that $\frac{dy}{dx} = 0$ and $y = 0$ when $x = 0$. ← Sections 7.3, 7.4

Population levels of predators and their prey can be modelled using a pair of coupled first-order differential equations. → Mixed exercise Q16

## 8.1 Modelling with first-order differential equations

First-order differential equations can be used to model problems in kinematics. You have covered some of these contexts in A level mathematics and you have learned the following relationships between displacement, velocity and acceleration:

$$v = \frac{ds}{dt} \qquad a = \frac{dv}{dt}$$

where $s$ is the displacement of a particle, $v$ is the velocity and $a$ is the acceleration at time $t$.

You can use these relationships to construct first-order differential equations which you can solve using the standard methods of direct integration and separation of variables from A level mathematics, or the method of integrating factors covered in Chapter 7 of this book.

### Example 1

A particle $P$ starts from rest at a point $O$ and moves along a straight line. At time $t$ seconds the acceleration, $a\ \text{m s}^{-2}$, of $P$ is given by

$$a = \frac{6}{(t+2)^2},\ t \geqslant 0$$

**a** Find the velocity of $P$ at time $t$ seconds.

**b** Show that the displacement of $P$ from $O$ when $t = 6$ is $(18 - 12\ln 2)$ m.

**a** $\dfrac{dv}{dt} = \dfrac{6}{(t+2)^2}$

Write the given relationship as a first-order differential equation.

$v = \displaystyle\int \frac{6}{(t+2)^2}\,dt = \int 6(t+2)^{-2}\,dt$

Integrate both sides with respect to $t$. Notice that this equation can be solved by direct integration.

$v = \dfrac{6(t+2)^{-1}}{-1} + c$

$= c - \dfrac{6}{t+2}$

Remember to include the constant of integration.

When $t = 0$, $v = 0$:

$0 = c - \frac{6}{2} \Rightarrow c = 3$

Use the initial conditions to find the value of $c$.

The velocity of $P$ at time $t$ seconds is

$\left(3 - \dfrac{6}{t+2}\right)\text{m s}^{-1}$.

**b** $\dfrac{ds}{dt} = 3 - \dfrac{6}{t+2}$

Set up another first-order differential equation using $\dfrac{ds}{dt} = v$.

$s = \displaystyle\int \left(3 - \frac{6}{t+2}\right)dt$

$= 3t - 6\ln(t+2) + d$

Solve this differential equation, again using standard methods, and include a second constant of integration. You are told that the particle starts from rest at the origin so your initial conditions are $s = 0$ when $t = 0$.

When $t = 0$, $s = 0$:

$0 = -6\ln 2 + d \Rightarrow d = 6\ln 2$

$s = 3t - 6\ln(t+2) + 6\ln 2$

When $t = 6$,

$s = 18 - 6\ln 8 + 6\ln 2$

$= 18 - 6\ln\left(\frac{8}{2}\right) = 18 - 6\ln 4$

$= 18 - 12\ln 2$

The displacement of $P$ from $O$ when $t = 6$ is $(18 - 12\ln 2)$ m, as required.

Use the laws of logarithms to simplify your answer into the form asked for in the question. This can be done in more than one way. The working shown here uses

$\ln 8 - \ln 2 = \ln\left(\frac{8}{2}\right) = \ln 4$

and

$\ln 4 = \ln 2^2 = 2\ln 2$.

## Example 2

A particle $P$ is moving along a straight line. At time $t$ seconds, the acceleration of the particle is given by

$$a = t + \frac{3}{t}v, \ t \geqslant 0$$

Given that $v = 0$ when $t = 2$, show that the velocity of the particle at time $t$ is given by the equation

$$v = ct^3 - t^2$$

where $c$ is a constant to be found.

$\frac{dv}{dt} - \frac{3}{t}v = t$

Integrating factor $= e^{\int P\,dt} = e^{\int -\frac{3}{t}\,dt}$

$= e^{-3\ln t}$

$= e^{\ln t^{-3}}$

$= t^{-3}$

So the original equation becomes

$t^{-3}\frac{dv}{dt} - 3t^{-4}v = t^{-2}$

$\Rightarrow \quad \frac{d}{dt}(t^{-3}v) = t^{-2}$

$\Rightarrow \quad t^{-3}v = -t^{-1} + c$

$\Rightarrow v = -t^2 + ct^3$

Since $v = 0$ when $t = 2$,

$0 = -4 + 8c \Rightarrow c = \frac{1}{2}$

So the velocity of the particle at time $t$ is given by $v = \frac{1}{2}t^3 - t^2$

Write the differential equation in the form $\frac{dv}{dt} + Pv = Q$ where $P$ and $Q$ are functions of $t$. To solve this differential equation, you need to use the integrating factor method. ← Section 7.1

Find the integrating factor and use the standard method to find the general solution.

Multiply through by $t^3$ to find $v$.

First-order differential equations can also be used to model other real-life situations involving rates of change.

You might need to construct or analyse models based on situations other than kinematics in your exam.

## Example 3

A storage tank initially contains 1000 litres of pure water. Liquid is removed from the tank at a constant rate of 30 litres per hour and a chemical solution is added to the tank at a constant rate of 40 litres per hour. The chemical solution contains 4 grams of copper sulphate per litre of water.

Given that there are $x$ grams of copper sulphate in the tank after $t$ hours and that the copper sulphate immediately disperses throughout the tank on entry,

**a** show that the situation can be modelled by the differential equation

$$\frac{dx}{dt} = 160 - \frac{3x}{100 + t}, \ t \geqslant 0$$

**b** Hence find the number of grams of copper sulphate in the tank after 6 hours.

**c** Explain how the model could be refined.

**a** Litres of liquid in the tank after $t$ hours is given by

$1000 + 40t - 30t = 1000 + 10t$

Let $x$ grams be the amount of copper sulphate in the tank after $t$ hours.

Concentration of copper sulphate after $t$ hours $= \frac{x}{1000 + 10t}$ grams per litre.

Rate copper sulphate in $= 40 \times 4$

$= 160$ grams per hour

Rate copper sulphate out $= 30 \times \frac{x}{1000 + 10t}$

$= \frac{3x}{100 + t}$ grams per hour

hence $\frac{dx}{dt} = 160 - \frac{3x}{100 + t}$

Write down the amount of liquid in the tank after $t$ hours.

**Problem-solving**

Work out the rate that copper sulphate enters the tank and leaves it. The rate at which the copper sulphate leaves the tank will be dependent on the concentration of copper sulphate in the tank.

Write down the final differential equation. $\frac{dx}{dt}$ is the rate of change of copper sulphate in the tank, so it is equal to the rate of copper sulphate in **minus** the rate of copper sulphate out.

**b** $\frac{dx}{dt} + \frac{3}{100 + t}x = 160$

Integrating factor $= e^{\int P dt} = e^{\int \frac{3}{100+t} dt}$

$= e^{3\ln(100+t)}$

$= e^{\ln(100+t)^3}$

$= (100 + t)^3$

Write the differential equation in the form $\frac{dx}{dt} + Px = Q$.

So the original equation becomes

$(100 + t)^3\frac{dx}{dt} + 3(100 + t)^2x = 160(100 + t)^3$

$\Rightarrow \quad \frac{d}{dt}((100 + t)^3 x) = 160(100 + t)^3$

$\Rightarrow \quad (100 + t)^3 x = 40(100 + t)^4 + c$

When $t = 0$, $x = 0$,

$0 = 4000000000 + c \Rightarrow c = -4 \times 10^9$

Work out the value of $c$ from the initial conditions.

After 6 hours,

$(106)^3 x = 40 \times (106)^4 - 4 \times 10^9$

$\Rightarrow x = 882$ g (3 s.f.)

c The model could be refined to take into account the fact that the copper sulphate does not disperse immediately on entering the tank.

You could give any sensible suggestion here, but make sure it is in the context of the model. For example, the rate at which liquid is removed could be made to vary with the volume of liquid in the tank.

## Exercise 8A

**1** A particle $P$ is moving along the $x$-axis in the direction of $x$ increasing. At time $t$ seconds, the velocity of $P$ is $(t \sin t)\,\text{m s}^{-1}$. When $t = 0$, $P$ is at the origin. Show that when $t = \frac{\pi}{2}$, $P$ is 1 metre from $O$.

**2** A particle $P$ is moving along a straight line. Initially $P$ is at rest. At time $t$ seconds $P$ has velocity $v\,\text{m s}^{-1}$ and acceleration $a\,\text{m s}^{-2}$ where

$$a = \frac{6t}{(2 + t^2)^2},\ t \geqslant 0$$

Find $v$ in terms of $t$.

(P) **3** A particle $P$ is moving along the $x$-axis. At time $t$ seconds $P$ has velocity $v\,\text{m s}^{-1}$ in the direction $x$ increasing and an acceleration of magnitude $4e^{0.2t}\,\text{m s}^{-2}$ in the direction $x$ decreasing. When $t = 0$, $P$ is moving through the origin with velocity $20\,\text{m}^{-1}$ in the direction $x$ increasing. Find:

**a** $v$ in terms of $t$

**b** the maximum value of $x$ attained by $P$ during its motion.

(P) **4** A particle, $P$, is moving along the $x$-axis. At time $t$ seconds, the displacement of the particle from $O$ is $x$ m, and its velocity is $v\,\text{m s}^{-1}$, where

$$v = e^{-\frac{x}{2}}$$

Given that the particle is initially at the origin, find $x$ as a function of $t$.

(E/P) **5** A sports car moves along a horizontal straight road. At time $t$ seconds the acceleration, in $\text{m s}^{-2}$, is modelled using the differential equation

$$\frac{dv}{dt} - 2vt = t$$

where $v$ is the velocity of the car in $\text{m s}^{-1}$.

When $t = 0$, the car is travelling at $1\,\text{m s}^{-1}$.

**a** Show that the velocity of the car at time $t$ seconds can be written as $v = \frac{1}{2}(3e^{t^2} - 1)$. **(5 marks)**

**b** Find the velocity of the car after 2 seconds. **(2 marks)**

**c** State, with a reason, whether the model is suitable when $t = 4$ seconds. **(2 marks)**

(E/P) **6** A raindrop falls vertically from rest through mist. Water condenses on the raindrop as it falls. You are given that the motion of the raindrop may be modelled by the equation

$$(t + 4)\frac{dv}{dt} + 4v = 9.8(t + 4)$$

where $t$ is the time in seconds and $v$ is the velocity of the raindrop in $\text{m s}^{-1}$.

**a** Show that $v = \dfrac{49(t + 4)^5 - c}{25(t + 4)^4}$ where $c$ is a constant to be found. **(6 marks)**

**b** Find the velocity of the raindrop after 5 seconds. **(2 marks)**

**c** By considering the velocity for large values of $t$, suggest one criticism of the model. **(1 mark)**

E/P **7** A gas storage tank initially contains 500 cm³ of helium. The helium leaks out at a constant rate of 20 cm³ per hour and a gas mixture is added to the tank at a constant rate of 50 cm³ per hour. The gas mixture contains 5% oxygen and 95% helium.

Given that there is $x$ cm³ of oxygen in the tank after $t$ hours and that the oxygen immediately mixes throughout the tank on entry,

**a** show that the situation can be modelled by the differential equation

$$\frac{dx}{dt} = 2.5 - \frac{2x}{50 + 3t}$$ **(4 marks)**

**b** Hence find the volume of oxygen in the tank after 4 hours. **(5 marks)**

**c** Explain how the model could be refined. **(1 mark)**

## 8.2 Simple harmonic motion

You can use second – order differential equations to model particles moving with **simple harmonic motion**.

- **Simple harmonic motion (S.H.M.) is motion in which the acceleration of a particle $P$ is always towards a fixed point $O$ on the line of motion of $P$. The acceleration is proportional to the displacement of $P$ from $O$.**

The point $O$ is called the **centre of oscillation**.

We write $\ddot{x} = -\omega^2 x$ — The minus sign means that the acceleration is always directed towards $O$.

This can be shown on a diagram




Given that $\ddot{x} = \frac{dv}{dt}$ you can use the chain rule to derive a relationship between $x$, $v$ and $a$ which will allow you to solve the second-order differential equation for simple harmonic motion:

$$\ddot{x} = \frac{dv}{dt} = \frac{dv}{dx} \times \frac{dx}{dt}$$

Since $\frac{dx}{dt} = v$, you can write acceleration as follows:

- $\ddot{x} = v\frac{dv}{dx}$

**Links** The constant of proportionality in this equation is $\omega^2$, where $\omega$ is the **angular velocity** of the particle. → FM2, Chapter 1

**Online** Explore simple harmonic motion using GeoGebra.



**Notation** You can use 'dot' notation to indicate differentiation with respect to time. $\dot{x} = \frac{dx}{dt}$ and $\ddot{x} = \frac{d^2x}{dt^2}$. So if $x$ is used to denote displacement, then $\dot{x}$ represents velocity and $\ddot{x}$ represents acceleration.

### Example 4

A particle $P$ moves with simple harmonic motion about a point $O$. Given that the maximum displacement of the particle from $O$ is $a$,

**a** show that $v^2 = \omega^2(a^2 - x^2)$, where $v$ is the velocity of the particle and $\omega^2$ is a constant.

**b** show that $x = a\sin(\omega t + \alpha)$, where $\alpha$ is an arbitrary constant.

**a** $\ddot{x} = -\omega^2 x$ and $\ddot{x} = v\frac{dv}{dx}$

So $v\frac{dv}{dx} = -\omega^2 x$

$\int v\,dv = -\int \omega^2 x\,dx$

$\frac{1}{2}v^2 = -\frac{1}{2}\omega^2 x^2 + c$

$v = 0$ when $x = a$,

therefore $0 = -\frac{1}{2}\omega^2 a^2 + c \Rightarrow c = \frac{1}{2}\omega^2 a^2$

Hence $\frac{1}{2}v^2 = -\frac{1}{2}\omega^2 x^2 + \frac{1}{2}\omega^2 a^2$

$\Rightarrow \quad v^2 = \omega^2(a^2 - x^2)$ as required.

**b** $\ddot{x} + \omega^2 x = 0$

$\frac{d^2x}{dt^2} + \omega^2 x = 0$

$m^2 + \omega^2 = 0 \Rightarrow m = \pm\omega i$

$x = A\cos\omega t + \beta\sin\omega t$

$= R\sin(\omega t + \alpha)$

This has maximum amplitude $R$. Since the maximum amplitude of the particle is $a$:

$x = a\sin(\omega t + \alpha)$ as required.

Write down the equation for simple harmonic motion and use $\ddot{x} = v\frac{dv}{dx}$ to write a differential equation involving $v$ and $x$ only.

Separate the variables.

Integrate both sides and add a constant of integration.

The velocity of the particle will be zero when the particle is at maximum displacement.

This is a second-order linear differential equation. Write it in the form $a\frac{d^2x}{dt^2} + b\frac{dx}{dt} + cx = 0$ and then solve the corresponding auxiliary equation.

← Secton 7.2

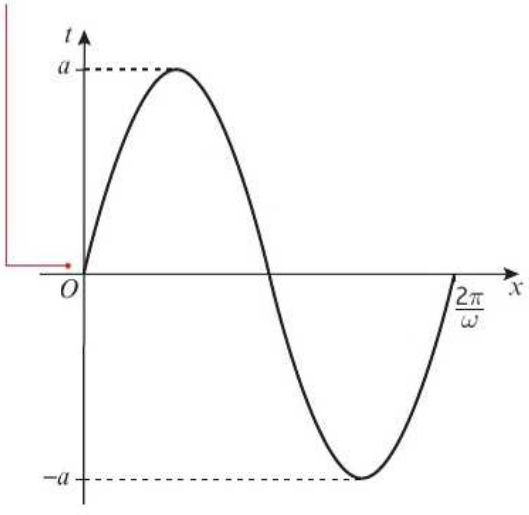
**Problem-solving**

You are given information about the maximum displacement, so write the general solution in a form where you can determine the amplitude. The arbitrary constants become $R$ and $\alpha$, where $R^2 = A^2 + B^2$ and $\tan\alpha = \frac{A}{B}$

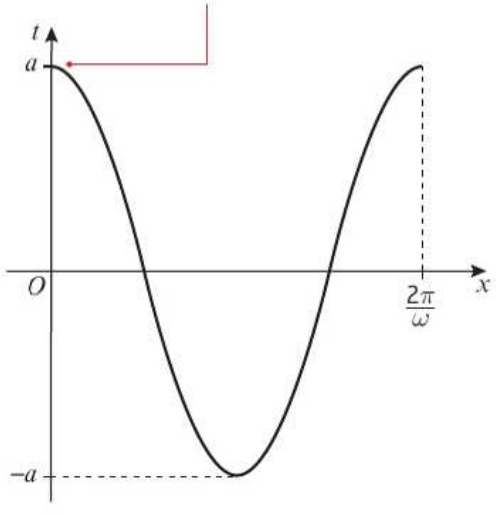
You can see from the above example that the path of a particle $P$ under simple harmonic motion is a sine wave with period $\frac{2\pi}{\omega}$ and amplitude $a$.

**Notation** The **period** of the motion is the time the particle takes to complete one oscillation. The **amplitude** of the motion is the maximum displacement from the origin.

If the particle is at its equilibrium position at time $t = 0$, then $\alpha = 0$.




If the particle is at its stationary point, i.e. its maximum displacement, when $t = 0$, then $\alpha = \frac{\pi}{2}$.



## Example 5

A particle is moving along a straight line. At time $t$ seconds its displacement, $x$ m from a fixed point $O$ is such that $\frac{d^2x}{dt^2} = -4x$.

Given that at $t = 0$, $x = 1$ and the particle is moving with velocity $4\,\text{m s}^{-1}$,

**a** find an expression for the displacement of the particle after $t$ seconds

**b** hence determine the maximum displacement of the particle from $O$.

**a** $\frac{d^2x}{dt^2} + 4x = 0$

$m^2 + 4 = 0 \Rightarrow m = \pm 2i$

$x = A\cos 2t + B\sin 2t$

$x = 1$ when $t = 0 \Rightarrow A = 1$

$v = \frac{dx}{dt} = -2\sin 2t + 2B\cos 2t$

$v = 4$ when $t = 0 \Rightarrow B = 2$

Hence $x = \cos 2t + 2\sin 2t$

**b** $x = \sqrt{5}\sin(2t + 0.4636...)$

Hence maximum displacement is $\sqrt{5}$ metres.

This is a second-order linear differential equation. Write it in the form $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + c = 0$.

Write down and solve the auxiliary equation.

Write down the general solution. There are two purely imaginary roots so the general solution is in the form $y = A\cos qx + B\sin qx$ ← Section 7.2

Substitute the initial condition for $x$: $1 = A\cos(0) + B\sin(0)$

Differentiate $x$ to find an expression for $v$ and use the initial condition for $v$: $4 = -2\sin(0) + 2B\cos(0)$

Write your solution to **a** in the form $R\sin(\theta + \alpha)$.

## Example 6

A particle, $P$, is attached to the ends of two identical elastic springs. The free ends of the springs are attached to two points $A$ and $B$. The point $C$ lies between $A$ and $B$ such that $ABC$ is a straight line and $AC \neq BC$. The particle is held at $C$ and then released from rest.

At time $t$ seconds, the displacement of the particle from $C$ is $x$ m and its velocity is $v\,\text{m s}^{-1}$.

The subsequent motion of the particle can be described by the differential equation $\ddot{x} = -25x$.

**a** Describe the motion of the particle.

Given that $x = 0.4$ and $v = 0$ when $t = 0$,

**b** solve the differential equation to find $x$ as a function of $t$

**c** state the period of the motion and calculate the maximum speed of $P$.

**a** The particle moves with simple harmonic motion.

**b** $\ddot{x} + 25x = 0$

$m^2 + 25 = 0 \Rightarrow m = \pm 5i$

$x = A\cos 5t + B\sin 5t$

$x = 0.4$ when $t = 0 \Rightarrow A = 0.4$

$\dot{x} = -5A\sin 5t + 5B\cos 5t$

$v = 0$ when $t = 0 \Rightarrow B = 0$

Hence $x = 0.4\cos 5t$

Write down and solve the auxiliary equation. Then write down the general solution.

Find the values of $A$ and $B$.

c Period of motion is $\frac{2\pi}{5}$ seconds

$\dot{x} = -2\sin 5t$, so maximum speed $= 2\,\text{m s}^{-1}$.

## Exercise 8B

(E) **1** A particle is moving along a straight line. At time $t$ seconds its displacement, $x$ m, from a fixed point $O$ is such that $\frac{d^2x}{dt^2} = -9x$.

**a** Describe the motion of the particle. **(1 mark)**

Given that at $t = 0$, $x = 2$ and the particle is moving with velocity $3\,\text{m s}^{-1}$,

**b** find an expression for the displacement of the particle after $t$ seconds. **(7 marks)**

**c** Hence determine the maximum distance of the particle from $O$. **(2 marks)**

(E) **2** A particle is moving with simple harmonic motion. After $t$ seconds its displacement, $x$ m from a fixed point $O$ is such that $\ddot{x} = -16x$.

Given that at $t = 0$, $x = 5$ and $\dot{x} = 2$,

**a** find an expression for the displacement of the particle after $t$ seconds. **(7 marks)**

**b** Hence determine the period of motion and the maximum distance of the particle from $O$. **(3 marks)**

(E/P) **3** A particle moves along a straight line. The particle moves such that its acceleration, in $\text{m s}^{-2}$, acts towards a fixed point $O$ and is proportional to its distance, $x$ m, from $O$.

**a** Describe the motion of the particle. **(1 mark)**

Given that the acceleration of the particle is $-5\,\text{m s}^{-2}$ when $x = 1$,

**b** write down a differential equation to describe the motion of the particle. **(2 marks)**

If the velocity and displacement of the particle at time $t = 0$ are $6\,\text{m s}^{-1}$ and 5 m respectively,

**c** find an expression for the displacement of the particle after $t$ seconds. **(7 marks)**

**d** Hence find the maximum distance of the particle from $O$. **(2 marks)**

(E/P) **4** A particle moves along a straight line such that its acceleration, in $\text{m s}^{-2}$, acts towards a fixed point $O$ and is proportional to its distance, $x$ m, from $O$.

The equation of motion is given as $\frac{d^2x}{dt^2} = -kx$, $t \geqslant 0$

where $k$ is a constant.

Given that the acceleration is $-7\,\text{m s}^{-2}$ when $x = 2$,

**a** write down the value of $k$. **(1 mark)**

**b** Use your answer to part **a** to find $x$ as a function of $t$ given that $x = 6$ and $\dot{x} = 1$ at time $t = 0$. **(7 marks)**

**c** Hence find, correct to 2 decimal places, the period of the motion. **(2 marks)**

E/P **5** A small rowing boat is floating on the surface of the sea, tied to a pier. The boat moves up and down in a vertical line, such that the vertical displacement, $x$ m, from its equilibrium point satisfies the equation

$$\frac{d^2x}{dt^2} = -2.25x,\ t \geqslant 0$$

Given that the maximum displacement is 1.3 m and it occurs when $t = 2$ seconds,

**a** find an expression for $x$ in terms of $t$ in the form $x = A\sin\omega t + B\cos\omega t$ where $A$, $B$ and $\omega$ are constants to be found. Give $A$ and $B$ correct to three decimal places. **(7 marks)**

**b** State the time elapsed between the boat being at its highest and lowest points. **(2 marks)**

**c** Criticise the model in terms of the motion of the boat for large values of $t$. **(1 mark)**

E/P **6** A particle $P$ is attached to one end of a light elastic spring. The other end of the spring is fixed to a point $A$ on the smooth horizontal surface on which $P$ rests. The particle is held at rest with $AP = 0.9$ m and then released. At time $t$ seconds the displacement of the particle from $A$ is $x$ m. The motion of the particle can be modelled using the equation $\ddot{x} = -200x$.

**a** State the type of motion exhibited by the particle $P$. **(1 mark)**

Given that $x = 0.3$ and the particle is at rest when $t = 0$,

**b** solve the differential equation to find $x$ as a function of $t$ **(7 marks)**

**c** find the period and amplitude of the motion **(3 marks)**

**d** calculate the maximum speed of $P$. **(2 marks)**

E **7** A particle $P$ is attached to one end of a light elastic spring. The other end of the spring is fixed to a point $O$ on the smooth horizontal surface on which $P$ rests with $OP = 2.6$ m and then released. The motion of $P$ can be described using the equation $\ddot{x} = -\frac{100}{0.64}x$, where $x$ m is the displacement of $P$ from $O$ at time $t$ seconds.

Given that $x = 1$ and the particle is at rest at time $t = 0$,

**a** solve the differential equation to find $x$ as a function of $t$ **(7 marks)**

**b** find the period of motion. **(2 marks)**

E **8** A smooth cylinder is fixed with its axis horizontal. A piston of mass 2.5 kg is inside the cylinder, attached to one end of the cylinder by a spring of natural length 50 cm. The piston is held at rest in the cylinder with the spring compressed to a length of 42 cm. The piston is then released. The spring can be modelled as a light elastic spring and the piston can be modelled as a particle.

The motion of $P$ can be described using the equation $\ddot{x} = -320x$, where $x$ cm is the compression of the spring from its natural length at time $t$ seconds.

Given that $x = 8$ and $v = 0$ when $t = 0$,

**a** solve the differential equation to find $x$ as a function of $t$ **(7 marks)**

**b** find the period of the resulting oscillations. **(2 marks)**

E/P **9** A pendulum $P$ is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point $A$ on a ceiling. The pendulum hangs in equilibrium at a point $B$ vertically below $A$.

The pendulum is then moved through a horizontal distance of 15 cm and released from rest.

The subsequent motion can be modelled using the equation $\ddot{x} = -\frac{250}{3}x$ where $x$ cm is the horizontal displacement from the vertical at time $t$ seconds and $t \geqslant 0$.

**a** Solve the differential equation to find $x$ as a function of $t$. **(7 marks)**

**b** Find the period and amplitude of the motion. **(3 marks)**

Anton says that since the pendulum describes simple harmonic motion, the model suggests that the pendulum will stay swinging forever.

**c** Suggest a refinement of the model in light of this statement. **(1 mark)**

## 8.3 Damped and forced harmonic motion

You can refine the model for simple harmonic motion by adding an additional force which is proportional to the velocity of the particle. When this force acts so as to slow the particle down it is known as a **damping force**, and the motion of the particle is known as **damped harmonic motion**.

- **For a particle moving with damped harmonic motion**

$$\frac{d^2x}{dt^2} + k\frac{dx}{dt} + \omega^2 x = 0$$

**where $x$ is the displacement from a fixed point at time $t$, and $k$ and $\omega^2$ are positive constants.**

**Notation** You could also write this as $\ddot{x} + k\dot{x} + \omega^2 x = 0$.

**Links** This is an example of a second-order homogeneous differential equation. You can solve equations like this by considering the auxiliary equation. ← Section 7.2

There are three separate cases corresponding to the auxiliary equation having distinct real, equal or complex roots.

When $k^2 > 4\omega^2$ there are two distinct real roots for the auxiliary equation. This is known as **heavy damping**. In this case there will be no oscillations performed as the resistive force is large compared with the restoring force.

When $k^2 = 4\omega^2$ the auxiliary equation has equal roots. This is known as **critical damping**. Again there will be no oscillations performed.

When $k^2 < 4\omega^2$ the auxiliary equation has complex roots. This is known as **light damping** and is the only case where oscillations are seen. The amplitude of the oscillations will decrease exponentially over time.

For heavy and critical damping the exact nature of the motion will depend on the initial conditions given.

For light damping the period of the observed oscillations can be calculated.

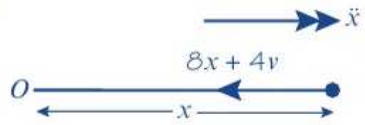
### Example 7

A particle $P$ of mass 0.5 kg moves in a horizontal straight line. At time $t$ seconds, the displacement of $P$ from a fixed point, $O$, on the line is $x$ m and the velocity of $P$ is $v$ m s$^{-1}$. A force of magnitude $8x$ N acts on $P$ in the direction $PO$. The particle is also subject to a resistance of magnitude $4v$ N.
When $t = 0$, $x = 1.5$ and $P$ is moving in the direction of increasing $x$ with speed 4 m s$^{-1}$.

**a** Show that $\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 16x = 0$

**b** Find the value of $x$ when $t = 1$.

a




$$F = ma$$
$$-(8x + 4v) = 0.5\ddot{x}$$
$$0.5\ddot{x} + 4\dot{x} + 8x = 0$$
$$\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 16x = 0$$

b Auxiliary equation:

$$m^2 + 8m + 16 = 0$$
$$(m + 4)^2 = 0$$
$$m = -4$$

General solution is $x = (A + Bt)e^{-4t}$

$t = 0, x = 1.5 \Rightarrow 1.5 = A$

$$v = \frac{dx}{dt} = Be^{-4t} - 4(A + Bt)e^{-4t}$$

$t = 0, v = 4 \Rightarrow 4 = B - 4A$

$$B = 4 + 4A = 4 + 4 \times 1.5 = 10$$

So $x = (1.5 + 10t)e^{-4t}$

$t = 1 \Rightarrow x = 11.5e^{-4} = 0.2106\ldots$

When $t = 1$, $x = 0.211$ (3 s.f.)

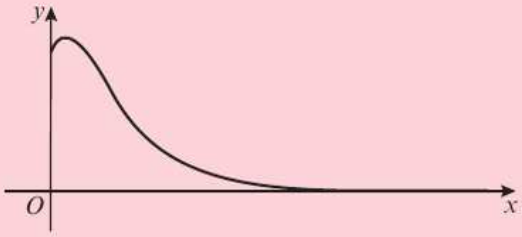
Draw a diagram to show the situation.

The auxiliary equation has equal roots. This means that the general solution is in the form $x = (A + Bt)e^{\alpha t}$, where $\alpha$ is the root. ← Section 7.2

Use the initial conditions given in the question to obtain values for $A$ and $B$.

**Problem-solving**

This is an example of **critical damping**:




**Online** Explore damped harmonic motion using GeoGebra.



## Example 8

A particle $P$ hangs freely in equilibrium attached to one end of a light elastic string. The other end of the string is attached to a fixed point $A$. The particle is now pulled down and held at rest in a container of liquid which exerts a resistance to motion on $P$. $P$ is then released from rest. While the string remains taut and the particle in the liquid, the motion can be modelled using the equation

$$\frac{d^2x}{dt^2} + 6k\frac{dx}{dt} + 5k^2x = 0, \text{ where } k \text{ is a positive real constant}$$

Find the general solution to the differential equation and state the type of damping that the particle is subject to.

Auxiliary equation: $m^2 + 6km + 5k^2 = 0$

$$(m + 5k)(m + k) = 0$$

$m = -5k$ or $-k$

General solution is $x = Ae^{-5kt} + Be^{-kt}$

The auxiliary equation has two distinct real roots so the particle is subject to heavy damping.

If the auxiliary equation has distinct real roots $\alpha$ and $\beta$, then the general solution is in the form $x = Ae^{\alpha t} + Be^{\beta t}$

## Example 9

One end of a light elastic spring is attached to a fixed point $A$. A particle $P$ is attached to the other end and hangs in equilibrium vertically below $A$. The particle is pulled vertically down from its equilibrium position and released from rest. A resistance proportional to the speed of $P$ acts on $P$. The equation of motion of $P$ is given as

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + 2k^2x = 0$$

where $k$ is a positive real constant and $x$ is the displacement of $P$ from its equilibrium position.

**a** Find the general solution to the differential equation.

**b** Write down the period of oscillation in terms of $k$.

**a** Auxiliary equation: $m^2 + 2km + 2k^2 = 0$

$$m = \frac{-2k \pm \sqrt{4k^2 - 4 \times 2k^2}}{2} = -k \pm ik$$

So $x = e^{-kt}(A\cos kt + B\sin kt)$

**b** Period $= \frac{2\pi}{k}$

If the auxiliary equation has complex roots $p \pm qi$, then the general solution is in the form $x = e^{pt}(A\cos qt + B\sin qt)$

$(A\cos kt + B\sin kt)$ can be written as $R\cos(kt + e)$ to give a period of $\frac{2\pi}{k}$

**Problem-solving**

This is an example of **light damping**:




The displacement of the particle will oscillate with a reducing amplitude.

You can investigate the motion of a particle which is subject to the same two forces as above but is also forced to oscillate with a frequency other than its natural one. This type of motion is called **forced harmonic motion**.

- **For a particle moving with forced harmonic motion**

$$\frac{d^2x}{dt^2} + k\frac{dx}{dt} + \omega^2x = f(t)$$

**where $x$ is the displacement from a fixed point at time $t$, and $k$ and $\omega^2$ are positive constants.**

**Links** This is an example of a second-order non-homogeneous differential equation. You will need to find the solution to the corresponding homogeneous equation, then add a **particular integral**, the form of which will depend on f($t$).

← Section 7.3

## Example 10

A particle $P$ of mass 1.5 kg is moving on the $x$-axis. At time $t$ the displacement of $P$ from the origin $O$ is $x$ metres and the speed of $P$ is $v\,\text{m s}^{-1}$. Three forces act on $P$, namely a restoring force of magnitude $7.5x$ N, a resistance to the motion of $P$ of magnitude $6v$ N and a force of magnitude $12\sin t$ N acting in the direction $OP$. When $t = 0$, $x = 5$ and $\frac{dx}{dt} = 2$.

**a** Show that $\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 8\sin t$.

**b** Find $x$ as a function of $t$.

**c** Describe the motion when $t$ is large.

**a**

$$F = ma$$

$$-7.5x - 6\frac{dx}{dt} + 12\sin t = 1.5\frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} + 5x + 4\frac{dx}{dt} = 8\sin t$$

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 8\sin t$$

**b** Auxiliary equation: $m^2 + 4m + 5 = 0$

$$m = \frac{-4 \pm \sqrt{4^2 - 4 \times 5}}{2} = \frac{-4 \pm i\sqrt{4}}{2} = -2 \pm i$$

Complementary function: $x_c = e^{-2t}(A\cos t + B\sin t)$

This is the general solution to the corresponding homogeneous differential equation.

Particular integral: try $x = p\sin t + q\cos t$

$$\frac{dx}{dt} = p\cos t - q\sin t$$

$$\frac{d^2x}{dt^2} = -p\sin t - q\cos t$$

So $(-p\sin t - q\cos t) + 4(p\cos t - q\sin t) + 5(p\sin t + q\cos t) = 8\sin t$

$$(-p - 4q + 5p)\sin t + (-q + 4p + 5q)\cos t = 8\sin t$$

Equating coefficients of $\cos t$: $4p + 4q = 0 \Rightarrow p + q = 0$

Equating coefficients of $\sin t$: $-4q + 4p = 8 \Rightarrow p - q = 2$

So $p = 1$ and $q = -1$

So the particular integral is $x = \sin t - \cos t$ and the general solution is $x = e^{-2t}(A\cos t + B\sin t) + \sin t - \cos t$

General solution = complementary function + particular integral

$t = 0$, $x = 5 \Rightarrow 5 = A - 1$ so $A = 6$

Use the initial conditions given in the question to obtain values for $A$ and $B$.

$x = e^{-2t}(A\cos t + B\sin t) + \sin t - \cos t$

$$\frac{dx}{dt} = -2e^{-2t}(A\cos t + B\sin t) + e^{-2t}(-A\sin t + B\cos t) + \cos t + \sin t$$

$t = 0$, $\frac{dx}{dt} = 2 \Rightarrow 2 = -2A + B + 1$ so $B = 2 + 12 - 1 = 13$

So $x = e^{-2t}(6\cos t + 13\sin t) + \sin t - \cos t$

**c** As $t \to \infty$, $e^{-2t} \to 0$

So $x \to \sin t - \cos t = R\sin(t - \alpha)$

Write $\sin t - \cos t$ in the form $R\sin(\theta - \alpha)$

$R = \sqrt{1 + 1} = \sqrt{2}$,

$\tan\alpha = 1 \Rightarrow \alpha = \frac{\pi}{2}$

$$x = \sqrt{2}\sin\left(t - \frac{\pi}{4}\right)$$

For large values of $t$, the motion is S.H.M. with amplitude $\sqrt{2}$ and period $2\pi$.

**Online** Explore forced harmonic motion using GeoGebra.



## Example 11

A particle $P$ is attached to end $A$ of a light elastic string $AB$. Initially the particle and the string lie at rest on a smooth horizontal plane. At time $t = 0$, the end $B$ of the string is set in motion and moves with constant speed $U$ in the direction $AB$, and the displacement of $P$ from $A$ is $x$. Air resistance acting on $P$ is proportional to its speed. The subsequent motion can be modelled by the differential equation

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + k^2x = 2kU$$

Find an expression for $x$ in terms of $U$, $k$ and $t$.

Auxiliary equation: $m^2 + 2km + k^2 = 0$

$$(m + k)^2 = 0$$

$$m = -k$$

Complementary function: $x = (A + Bt)e^{-kt}$

Particular integral: try $x = a$

The right–hand side of the differential equation is constant, so try a constant for the particular integral.

$\dot{x} = \ddot{x} = 0$

So $k^2a = 2kU \Rightarrow a = \frac{2U}{k}$

General solution is $x = (A + Bt)e^{-kt} + \frac{2U}{k}$

$t = 0, \quad x = 0 \quad \Rightarrow 0 = A + \frac{2U}{k} \Rightarrow A = -\frac{2U}{k}$

$\dot{x} = -k(A + Bt)e^{-kt} + Be^{-kt}$

$t = 0 \Rightarrow \dot{x} = U$

Use the initial conditions given in the question to obtain values for $A$ and $B$.

$U = -kA + B$

$B = U + kA = -U$

So $x = \left(-\frac{2U}{k} - Ut\right)e^{-kt} + \frac{2U}{k}$

## Exercise 8C

**1** A particle $P$ is moving in a straight line. At time $t$, the displacement of $P$ from a fixed point on the line is $x$. The motion of the particle is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = 0$$

When $t = 0$, $P$ is at rest at the point where $x = 2$.

**a** Find $x$ as a function of $t$.

**b** Calculate the value of $x$ when $t = \frac{\pi}{3}$

**c** State whether the motion is heavily, critically or lightly damped.

**2** A particle $P$ is moving in a straight line. At time $t$, the displacement of $P$ from a fixed point on the line is $x$. The motion of the particle is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 12x = 0$$

When $t = 0$, $P$ is at rest at the point where $x = 4$.

Find $x$ as a function of $t$.

E/P **3** A particle $P$ is moving in a straight line. At time $t$, the displacement of $P$ from a fixed point on the line is $x$. The motion of the particle is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 6x = 0$$

When $t = 0$, $P$ is at rest at the point where $x = 1$.

**a** Find $x$ as a function of $t$. **(6 marks)**

The smallest value of $t$, $t > 0$, for which $P$ is instantaneously at rest is $T$.

**b** Find the value of $T$. **(2 marks)**

E/P **4** A particle $P$ is attached to one end of a light elastic spring. The other end of the spring is attached to a fixed point $A$ and $P$ hangs freely in equilibrium vertically below $A$. At time $t = 0$, $P$ is projected vertically downwards with speed $u$. A resistance proportional to the speed of $P$ acts on $P$. The motion of $P$ can be modelled using the differential equation

$$\frac{d^2x}{dt^2} + 4k\frac{dx}{dt} + 4k^2x = 0$$

where $x$ is the displacement of $P$ from its equilibrium position at time $t$ and $k$ is a positive constant.

**a** Find an expression for $x$ in terms of $u$, $t$ and $k$. **(6 marks)**

**b** Find the time at which $P$ comes to instantaneous rest. **(2 marks)**

E/P **5** A particle of mass 2 kg moves in a horizontal straight line. At time $t$ seconds, the displacement of $P$ from a fixed point $O$ is $x$ metres and the speed of $P$ is $v\,\text{m s}^{-1}$.

A force of magnitude $6x$ N acts on $P$ in the direction $PO$. The particle is also subject to a resistance of magnitude $2v$ N. When $t = 0$, $x = 1$ and $P$ is moving in the direction of increasing $x$ with speed $2\,\text{m s}^{-1}$.

**a** Show that $\ddot{x} + \dot{x} + 3x = 0$. **(2 marks)**

**b** Solve the differential equation in part **a** to find $x$ as a function of $t$. **(8 marks)**

**c** Find the value of $x$ when $t = 2$. **(2 marks)**

**d** Describe the motion of $P$ for large values of $t$. **(1 mark)**

E/P **6** A particle $P$ is attached to end $A$ of a light elastic spring $AB$. The end $B$ of the spring is oscillating. At time $t$ the displacement of $P$ from a fixed point is $x$. When $t = 0$, $x = 0$ and $\frac{dx}{dt} = \frac{k}{5}$ where $k$ is a constant. Given that $x$ satisfies the differential equation

$$\frac{d^2x}{dt^2} + 9x = k\cos t$$

find $x$ as a function of $t$. **(8 marks)**

E/P **7** A particle $P$ is attached to end $A$ of a light elastic spring $AB$. Initially the spring and the particle lie at rest on a horizontal surface. The end $B$ of the spring is then moved in a straight line in the direction $AB$ with constant speed $U$. As $P$ moves it is subject to a resistance proportional to its speed. The extension, $x$, in the spring can be modelled using the differential equation

$$\frac{d^2x}{dt^2} + 5k\frac{dx}{dt} + 6k^2x = 5kU$$

Find an expression in $x$ in terms of $t$. **(8 marks)**

E/P **8** An engineering student is designing an oscillating piston that is attached to a vertical rod. The piston is to be released from rest from a point half way up the rod so that it oscillates in a vertical line. The vertical displacement, $x$ metres, of the top of the piston below its initial position at time $t$ seconds is modelled by the differential equation,

$$2\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + x = 100\cos t,\ t \geqslant 0$$

**a** Show that a particular solution to the differential equation is $x = 30\sin t - 10\cos t$ **(3 marks)**

**b** Hence find the general solution to the differential equation. **(5 marks)**

**c** Use the model to find, to the nearest centimetre, the vertical distance of the top of the piston from its initial position 5 seconds after it is released. **(4 marks)**

## 8.4 Coupled first-order simultaneous differential equations

In some real-life situations rates of change of two variables are connected. For example, in a predator–prey model, the rate of change of the population of bears might be dependent on both the number of bears and the number of fish in a river. Simultaneously, the rate of change of the number of fish might be dependent on both the number of bears and the number of fish. In this case, you have two dependent variables, the number of bears and fish, and one independent variable, time. You can set up two first-order differential equations to model the rates of change of the numbers of bears and fish.

Letting the number of bears at time $t$ be $x$ and the number of fish at time $t$ be $y$, you can write:

$$\frac{dx}{dt} = ax + by + f(t)$$

$$\frac{dy}{dt} = cx + dy + g(t)$$

**Notation** If $f(t)$ and $g(t)$ are both zero, the system is said to be **homogenous**.

These equations are called **coupled first-order linear differential equations** and you can solve them simultaneously to find $x$ and $y$ as functions of $t$.

- **You can solve coupled first-order linear differential equations by eliminating one of the dependent variables to form a second-order differential equation.**

### Example 12

At the start of the year 2010, a survey began on the numbers of bears and fish on a remote island in Northern Canada. After $t$ years the number of bears, $x$, and the number of fish, $y$, on the island are modelled by the differential equations

$$\frac{dx}{dt} = 0.3x + 0.1y \qquad (1)$$

$$\frac{dy}{dt} = -0.1x + 0.5y \qquad (2)$$

**a** Show that $\frac{d^2x}{dt^2} - 0.8\frac{dx}{dt} + 0.16x = 0$.

**b** Find the general solution for the number of bears on the island at time $t$.

**c** Find the general solution for the number of fish on the island at time $t$.

**d** At the start of 2010 there were 5 bears and 20 fish on the island.
Use this information to find the number of bears predicted to be on the island in 2020.

**e** Comment in the suitability of the model.

**a** $y = 10\frac{dx}{dt} - 3x$

$\frac{dy}{dt} = 10\frac{d^2x}{dt^2} - 3\frac{dx}{dt}$

$10\frac{d^2x}{dt^2} - 3\frac{dx}{dt} = -0.1x + 0.5\left(10\frac{dx}{dt} - 3x\right)$

$10\frac{d^2x}{dt^2} - 8\frac{dx}{dt} + 1.6x = 0$

$\frac{d^2x}{dt^2} - 0.8\frac{dx}{dt} + 0.16x = 0$

**b** $m^2 - 0.8m + 0.16 = 0 \Rightarrow m = 0.4$

Hence $x = Ae^{0.4t} + Bte^{0.4t}$

**c** $\frac{dx}{dt} = 0.4Ae^{0.4t} + 0.4Bte^{0.4t} + Be^{0.4t}$

$y = 10\frac{dx}{dt} - 3x$ from equation (1)

$y = 10(0.4Ae^{0.4t} + 0.4Bte^{0.4t} + Be^{0.4t}) - 3(Ae^{0.4t} + Bte^{0.4t})$

$y = Ae^{0.4t} + 10Be^{0.4t} + Bte^{0.4t}$

**d** At $t = 0$, $x = 5 \Rightarrow A = 5$

At $t = 0$, $y = 20 \Rightarrow B = 1.5$

$x = 5e^{0.4t} + 1.5te^{0.4t}$

At time $t = 10$: $x = 5e^4 + 15e^4 = 1092$

The model predicts there will be 1092 bears on the island by 2020.

**e** The model predicts the number of bears (and the number of fish) will grow without limit so it is unlikely to be realistic.

Rearrange equation (1) to make $y$ the subject.

Differentiate both sides with respect to $t$.

Substitute your expressions for $y$ and $\frac{dy}{dt}$ into equation (2) to form a second-order differential equation in $x$ and $t$.

Rearrange into the correct form.

Write down the auxiliary equation and solve it.

Use the general form of the solution of a differential equation with a repeated root.

**Problem-solving**

You do not need to go through the whole process to find an expression for $y$ in terms of $t$. If you differentiate your answer to part **b** with respect to $t$ you will have expressions for $x$ and $\frac{dx}{dt}$ in terms of $t$ only. You can substitute these into equation (1) and simplify to obtain the answer.

Use the initial conditions to find $A$ and $B$.

Substitute $t = 10$ into the equation for $x$. Round to the nearest whole number.

## Example 13

Two barrels contain contaminated water. At time $t$ seconds, the amount of contaminant in barrel $A$ is $x$ ml and the amount of contaminant in barrel $B$ is $y$ ml. Additional contaminated water flows into barrel $A$ at a rate of 5 ml per second. Contaminated water flows from barrel $A$ to barrel $B$ and from barrel $B$ to barrel $A$ through two connecting hoses, and drains out of barrel $A$ to leave the system completely.

The system is modelled using the differential equations

$$\frac{dx}{dt} = 5 + \frac{4}{9}y - \frac{1}{7}x \qquad (1)$$

$$\frac{dy}{dt} = \frac{3}{70}x - \frac{4}{9}y \qquad (2)$$

Show that $630\frac{d^2y}{dt^2} + 370\frac{dy}{dt} + 28y = 135$.

$$\frac{dy}{dt} + \frac{4}{9}y = \frac{3}{70}x \text{ from equation (2)}$$

$$\frac{70}{3}\frac{dy}{dt} + \frac{280}{27}y = x \quad (3)$$

$$\frac{dx}{dt} = \frac{70}{3}\frac{d^2y}{dt^2} + \frac{280}{27}\frac{dy}{dt} \quad (4)$$

$$\frac{70}{3}\frac{d^2y}{dt^2} + \frac{280}{27}\frac{dy}{dt} = 5 + \frac{4}{9}y - \frac{1}{7}\left(\frac{70}{3}\frac{dy}{dt} + \frac{280}{27}y\right)$$

$$70\frac{d^2y}{dt^2} + \frac{280}{9}\frac{dy}{dt} = 15 + \frac{4}{3}y - 10\frac{dy}{dt} - \frac{40}{9}y$$

$$630\frac{d^2y}{dt^2} + 280\frac{dy}{dt} = 135 + 12y - 90\frac{dy}{dt} - 40y$$

$$\Rightarrow 630\frac{d^2y}{dt^2} + 370\frac{dy}{dt} + 28y = 135 \text{ as required.}$$

Rearrange (2) to make $x$ the subject.

Differentiate (3) to find $\frac{dx}{dt}$.

Substitute (3) and (4) into (1), simplify coefficients and remove fractions.

Write your answer in the form required in the question.

## Exercise 8D

(P) **1** Find the particular solutions to the differential equations

$$\frac{dx}{dt} = x + y$$

$$\frac{dy}{dt} = x - y$$

given that $x = 1$ and $y = 2$ at $t = 0$.

(P) **2 a** Find the general solutions to the differential equations

$$\frac{dx}{dt} = x + 5y$$

$$\frac{dy}{dt} = -3y - x$$

**b** Given that at time $t = 0$, $x = 1$ and $y = 2$, find the particular solutions.

(P) **3** A system of differential equations is given as

$$\frac{dx}{dt} = 2x - 3y - 2$$

$$\frac{dy}{dt} = x + y - 1$$

Given that $x = 0$ and $y = 1$ when $t = 0$, find the particular solutions to the system of differential equations.

(E/P) **4** At the start of 2012, a survey began on the number of sand foxes and the number of meerkats on a remote desert island. After $t$ years, the number of sand foxes, $x$, and the number of meerkats, $y$, on the island are modelled by the differential equations

$$\frac{dx}{dt} = 0.2x + 0.2y$$

$$\frac{dy}{dt} = -0.5x + 0.4y$$

**a** Show that $\frac{d^2x}{dt^2} - 0.6\frac{dx}{dt} + 0.18x = 0$. **(3 marks)**

**b** Show that the general solution for the number of sand foxes at time $t$ is

$x = e^{\alpha t}(A\cos\beta t + B\sin\beta t)$

where $\alpha$ and $\beta$ are constants to be found. **(4 marks)**

**c** Find the general solution for the number of meerkats at time $t$.
Give your answer in the form $y = Pe^{\alpha t}(Q\cos\beta t + R\sin\beta t)$ where $P$ is a constant to be found and $Q$ and $R$ are functions of $A$ and $B$. **(3 marks)**

**d** Given that there were initially 3 sand foxes and 111 meerkats on the island, during which year does the model predict that the meerkats die out? **(5 marks)**

**e** How many sand foxes will there be when the meerkats die out? **(1 mark)**

**f** Use your answers to parts **d** and **e** to comment on the model. **(1 mark)**

E/P **5** A tank of water contains two different types of chemical that react with each other. The rates of change of each chemical can be modelled using the differential equations

$$\frac{dx}{dt} = -3x + 2y$$
$$\frac{dy}{dt} = -2x + y$$

where $x$ is the number of litres of chemical $X$ and $y$ is the number of litres of chemical $Y$ at time $t$ hours.

Initially there is one litre of chemical $X$ and two litres of chemical $Y$ in the tank.

**a** Show that the solutions to the differential equations can be written as

$x = Pe^{-t}$

$y = Qe^{-t}$

where $P$ and $Q$ are functions of $t$ to be found. **(8 marks)**

**b** Find, correct to three significant figures, the amount of each chemical at time $t = 2$ hours. **(2 marks)**

**c** Use the model to describe what happens to the amount of each chemical as $t$ gets large. **(2 marks)**

E/P **6** A freely hanging pendulum oscillates in both the $x$ and $y$ directions. At time $t$, the rates of change of the $x$ and $y$ displacements are given by the differential equations

$$\frac{dx}{dt} = -4y$$
$$\frac{dy}{dt} = 4x$$

**a** Show that the pendulum describes simple harmonic motion in the $y$ direction. **(3 marks)**

Given that the initial displacement of the pendulum is $x = 4$ and $y = 5$,

**b** find the particular solutions to the system of differential equations. **(6 marks)**

E/P **7** Joanna is investigating how a harmful substance from pollution is absorbed by the human body. She finds that the substance enters the body in the bloodstream and is transferred between the blood and the bodily organs. The harmful substance is then expelled from the body through, for example, sweat.

The amount of the substance in the blood, $x$ mg, and the organs, $y$ mg, at time $t$ days, can be modelled using the differential equations

$$\frac{dx}{dt} = -0.03x + 0.01y + 50$$

$$\frac{dy}{dt} = 0.01x - 0.03y$$

**a** Show that $\frac{d^2x}{dt^2} + 0.06\frac{dx}{dt} + 0.0008x = 1.5$ **(6 marks)**

**b** Hence find the general solutions to the system of differential equations in the form

$$x = f(t)$$
$$y = g(t)$$ **(8 marks)**

**c** Describe what happens to the amount of the substance in the blood and the organs as $t$ gets large. **(2 marks)**

E/P **8** A biologist is examining the rates of change of nutrients in both tree roots and the surrounding soil. Nutrients pass from the tree roots into the soil and from the soil into the tree roots. Nutrients also enter the system through both the roots and the soil and escape from the system in the same way.

The biologist believes that the amount of nutrients in the roots, $x$, and the amount of nutrients in the soil, $y$, at time t hours, can be modelled using the differential equations

$$\frac{dx}{dt} = -2x + y + 1$$

$$\frac{dy}{dt} = 4x + y + 2$$

**a** Show that $\frac{d^2x}{dt^2} + \frac{dx}{dt} - 6x = 1$. **(4 marks)**

**b** Find the general solutions to the system of differential equations. **(6 marks)**

**c** Using your answers to part **b**, comment on the suitability of the biologist's model. **(2 marks)**

**Challenge**

A closed environment supports populations of owls and field mice. At time $t$ months, the sizes of each population are $x$ and $y$ respectively. The situation is modelled by the pair of differential equations

$$\frac{dy}{dt} = 2y - \frac{y^2}{6000} - 60x$$

$$\frac{dx}{dt} = 0.02y - x$$

Find the number of owls and the number of field mice such that the population of both is stable.

**Hint** A population is stable when its rate of growth is zero.

## Mixed exercise 8

(E) **1** A particle $P$ moves along the $x$-axis in the direction of $x$ increasing. At time $t$ seconds, the velocity of $P$ is $v\,\text{m s}^{-1}$ and its acceleration is $20te^{-t^2}\,\text{m s}^{-2}$. When $t = 0$ the velocity of $P$ is $8\,\text{m s}^{-1}$. Find:

**a** $v$ in terms of $t$ **(4 marks)**

**b** the limiting velocity of $P$. **(3 marks)**

(E) **2** A particle $P$ moves along a straight line. Initially $P$ is at rest at a point $O$ on the line. At time $t$ seconds, where $t \geqslant 0$, the acceleration of $P$ is $\frac{18}{(2t+3)^3}\,\text{m s}^{-2}$ directed away from $O$.

Find the value of $t$ for which the speed of $P$ is $0.48\,\text{m s}^{-1}$. **(5 marks)**

(E) **3** A car moves along a horizontal straight road. At time $t$ seconds the acceleration of the car is $\frac{100}{(2t+5)^2}\,\text{m s}^{-2}$ in the direction of motion of the car. When $t = 0$, the car is at rest. Find:

**a** an expression for $v$ in terms of $t$ **(4 marks)**

**b** the distance moved by the car in the first 10 seconds of its motion. **(4 marks)**

(E) **4** A particle $P$ is moving in a straight line with acceleration $\cos^2 t\,\text{m s}^{-2}$ at time $t$ seconds. The particle is initially at rest at a point $O$.

**a** Find the speed of $P$ when $t = \pi$. **(4 marks)**

**b** Show that the distance of $P$ from $O$ when $t = \frac{\pi}{4}$ is $\frac{1}{64}(\pi^2 + 8)\,\text{m}$. **(4 marks)**

(E) **5** A particle $P$ is moving along the $x$-axis. At time $t$ seconds, $P$ has velocity $v\,\text{m s}^{-1}$ in the direction $x$ increasing and an acceleration of magnitude $\frac{2t+3}{t+1}\,\text{m s}^{-2}$ in the direction $x$ increasing.

When $t = 0$, $P$ is at rest at the origin $O$. Find:

**a** $v$ in terms of $t$ **(4 marks)**

**b** the distance of $P$ from $O$ when $t = 2$. **(4 marks)**

(E/P) **6** A colony of bacteria reproduces in a laboratory jar. At time $t = 0$, the volume of bacteria is $1\,\text{cm}^3$. Scientist Steve suggests that the rate of growth of the bacteria can be modelled using the differential equation

$$\frac{dV}{dt} = 2t + 3V + 5$$

where $t$ is the time in hours and $V$ is the volume in $\text{cm}^3$.

**a** Show that $V = At + B + Ce^{3t}$ where $A$, $B$ and $C$ are exact constants to be found. **(6 marks)**

**b** Find the volume of bacteria after 2 hours. **(2 marks)**

**c** Give one criticism of Steve's model and suggest one refinement he could make. **(2 marks)**

(E/P) **7** A fluid reservoir initially contains 10 000 litres of unpolluted fluid.

The reservoir is leaking at a constant rate of 200 litres per day.

It is suspected that contaminated fluid flows into the reservoir at a constant rate of 300 litres per day and that the contaminated fluid contains 4 grams of contaminant in every litre of fluid.

It is assumed that the contaminant instantly disperses throughout the reservoir upon entry.

Given that there are $x$ grams of contaminant in the reservoir after $t$ days,

**a** Show that the situation can be modelled by the differential equation

$$\frac{dx}{dt} = 1200 - \frac{2x}{100 + t}$$ **(4 marks)**

**b** Hence find, correct to three significant figures, the number of grams of contaminant in the reservoir after 7 days. **(5 marks)**

**c** Explain how the model could be refined. **(1 mark)**

(E) **8** A particle $P$ is attached to one end of a light elastic string of natural length 1.2 m. The other end of the string is attached to a fixed point $A$. The particle is hanging in equilibrium at the point $O$, which is vertically below $A$.

The particle is now displaced to a point $B$, vertically below $A$, and released from rest.

The subsequent motion while the string remains taut can be modelled using the equation $\ddot{x} = -49x$, where $x$ m is the displacement of the particle from $O$ at time $t$ seconds.

**a** Describe the motion of the particle while the string remains taut. **(1 mark)**

**b** Solve the differential equation and hence find the period of the motion. **(6 marks)**

(E) **9** A particle $P$ of mass 0.6 kg is attached to one end of a light elastic spring of natural length 2.5 m. The other end of the spring is attached to a fixed point $A$ on the smooth horizontal table on which $P$ lies. The particle is held at the point $B$ where $AB = 4$ m and released from rest.

The motion of the particle can be described using the equation $\ddot{x} = -\frac{50}{3}x$, where $x$ m is the displacement of the particle from $A$ at time $t$ seconds.

**a** Describe the motion of the particle. **(1 mark)**

**b** Solve the differential equation and hence find the period and amplitude of the motion. **(7 marks)**

(E/P) **10** A fisherman's float bobs up and down on the surface of the water. The float moves up and down in a vertical line, such that the vertical displacement, $x$ cm, from its equilibrium point satisfies the equation

$$\frac{d^2x}{dt^2} = -0.25x, \; t \geqslant 0$$

Given that the maximum displacement is 4 cm and it occurs when $t = 2$ seconds,

**a** find an expression for $x$ in terms of $t$ in the form $x = A\sin\omega t + B\cos\omega t$ where $A$, $B$ and $\omega$ are constants to be found. Give $A$ and $B$ correct to three decimal places. **(7 marks)**

**b** State the time elapsed between the float being at its highest and lowest points. **(2 marks)**

**c** Criticise the model in terms of the motion of the float. **(1 mark)**

(E/P) **11** A particle $P$ of mass $m$ is moving in a straight line. At time $t$ the displacement of $P$ from a fixed point $O$ on the line is $x$. Given that $x$ satisfies the differential equation

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + n^2x = 0$$

where $k$ and $n$ are positive constants with $k < n$,

**a** find an expression for $x$ in terms of $k$, $n$ and $t$. **(6 marks)**

**b** Write down the period of the motion. **(1 mark)**

E/P **12** A particle $P$ of mass $m$ is attached to one end of light elastic spring. The other end of the spring is attached to a fixed point $A$ and $P$ is hanging in equilibrium with $AP$ vertical.

The particle is now projected vertically downwards from its equilibrium position with speed $U$. A resistance of magnitude $2mkv$, where $v$ is the speed of $P$, acts on $P$. At time $t$, $t > 0$, the displacement of $P$ from its equilibrium position is $x$.

The motion of the particle is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + 2k^2x = 0$$

**a** Show that $P$ is instantaneously at rest when $kt = (n + \frac{1}{4})\pi$, where $n \in \mathbb{N}$. **(9 marks)**

**b** Sketch the graph of $x$ against $t$. **(3 marks)**

E/P **13** A particle $P$ of mass $m$ is attached to one end of a light elastic spring. The other end of the spring is attached to the roof of a stationary lift. The particle is hanging in equilibrium with the spring vertical. At time $t = 0$ the lift starts to move vertically upwards with constant speed $U$. At time $t$, $t > 0$, the displacement of $P$ from its initial position is $x$.

The motion of the particle is modelled by the differential equation

$$\frac{d^2x}{dt^2} + n^2x = n^2Ut$$

**a** find an expression for $x$ in terms of $t$ and $n$. **(8 marks)**

At time $t = T$, the particle is instantaneously at rest. Find:

**b** the smallest value of $T$ **(3 marks)**

**c** the displacement of $P$ from its initial position at this time. **(1 mark)**

E/P **14** A theme park ride designer is designing a new ride where the passengers will be in an enclosed car attached to a horizontal bar. The car will be released from rest from a point half way along the bar so that it oscillates in a horizontal line. The horizontal displacement, $x$ metres, of the centre of the car relative to its initial position at time $t$ seconds is modelled by the differential equation,

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 3x = 150\cos t,\ t \geqslant 0$$

**a** Show that a particular solution to the differential equation is

$$x = 30\sin t + 15\cos t$$ **(3 marks)**

**b** Hence find the general solution to the differential equation. **(5 marks)**

**c** Use the model to find, to the nearest metre, the horizontal distance of the centre of the car from its initial position 10 seconds after it is released. **(4 marks)**

E/P **15** **a** Find the general solution to the differential equation $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 10x = 27\cos t - 6\sin t$. **(8 marks)**

The equation is used to model water flow in a reservoir. At time $t$ days, the level of the water above a fixed level is $x$ m. When $t = 0$, $x = 3$ and the water level is rising at 6 metres per day.

**b** Find an expression for $x$ in terms of $t$. **(2 marks)**

**c** Show that after about a week, the difference between the lowest and highest water level is approximately 6 m. **(3 marks)**

E/P **16** At the start of 2008 a survey began on the number of hedgehogs and the number of slugs in a closed ecosystem. After $t$ years, the number of hedgehogs, $x$, and the number of slugs, $y$, in the ecosystem are modelled by the differential equations

$$\frac{dx}{dt} = 2x + y$$
$$\frac{dy}{dt} = -2x + 4y$$

**a** Show that $\frac{d^2x}{dt^2} - 6\frac{dx}{dt} + 10x = 0$. **(3 marks)**

**b** Show that the general solution for the number of hedgehogs at time $t$ is
$x = e^{\alpha t}(A\cos\beta t + B\sin\beta t)$
where $\alpha$ and $\beta$ are constants to be found. **(4 marks)**

**c** Find the general solution for the number of slugs at time $t$. **(3 marks)**

**d** Given that there were initially 10 hedgehogs and 20 slugs in the ecosystem, during which year does the model predict that the slugs die out? **(5 marks)**

**e** How many hedgehogs will there be when the slugs die out? **(1 mark)**

**f** Use your answers to parts **d** and **e** to comment on the model. **(1 mark)**

E/P **17** A sealed tank of bionutrient contains two different types of organism that interact with each other. The rates of change of each organism can be modelled using the differential equations

$$\frac{dx}{dt} = -4x + 3y$$
$$\frac{dy}{dt} = -3x + 2y$$

where $x$ is the number of organism $M$ and $y$ is the number of organism $N$ at time $t$ days. Initially $x = 10$ and $y = 20$.

**a** Show that the solutions to the differential equations can be written as
$x = Ate^{-t} + Be^{-t}$
$y = Cte^{-t} + De^{-t}$
where $A$, $B$, $C$ and $D$ are constants to be found. **(8 marks)**

**b** Find, correct to the nearest organism, the number of each organism at time $t = 2.1$ days. **(2 marks)**

**c** Use the model to describe what happens to the numbers of each type of organism as $t$ gets large. **(2 marks)**

**18** An industrial process consists of two linked tanks, $A$ and $B$, containing a chemical solution. The solution is free to pass between the tanks but it flows from $A$ to $B$ at a different rate than it flows from $B$ to $A$. The solution also enters both tanks, and flows directly out of tank $B$. The situation is modelled using the differential equations

$$\frac{dx}{dt} = 2 + \frac{1}{3}y - \frac{1}{2}x \qquad (1)$$
$$\frac{dy}{dt} = 1 + \frac{1}{2}x - \frac{2}{3}y \qquad (2)$$

where $x$ litres is the amount of solution in tank $A$ and $y$ litres is the amount of solution in tank $B$ at time $t$ minutes.

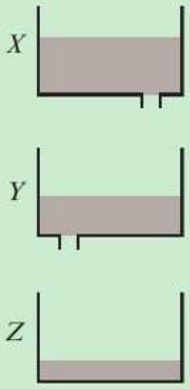
**a** Show that $6\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + y = 9$ **(6 marks)**

**b** Given that both tanks initially contain 8 litres of solution, find $x$ and $y$ as functions of $t$. **(7 marks)**

**c** State, with a reason, the approximate amount of solution in each tank after the system has been running for a long time. **(2 marks)**

## Challenge

**a** Three water tanks are positioned as shown in the diagram. Water flows from tank $X$ to tank $Y$, and from tank $Y$ to tank $Z$, by means of identical taps. Each tap allows water to flow at a rate of $r$ gallons/hour, where $r$ is the amount of water in the corresponding tank in gallons.
Initially, tank $X$ contains 300 gallons of water, tank $Y$ contains 200 gallons of water, and tank $Z$ contains 100 gallons of water. The taps are opened.

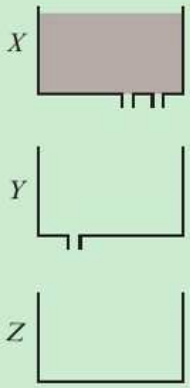



**i** Show that after $t$ hours have elapsed, the amount of water in tank $X$ is $300e^{-t}$ gallons.

**ii** Find the number of minutes after which tanks $X$ and $Y$ contain the same amount of water.

**iii** Find an expression for the amount of water in tank $Z$ after $t$ hours.

**b** A second identical tap is attached to tank $X$, which is filled to the brim. Tanks $Y$ and $Z$ are emptied, and all three taps are opened.




**i** Show that the amount of water in tank $Y$ is at a maximum after 42 minutes have elapsed, to the nearest minute.

**ii** Find the exact time elapsed before the amounts of water in tanks $Y$ and $Z$ are equal.

## Summary of key points

**1** **Simple harmonic motion** (S.H.M.) is motion in which the acceleration of a particle $P$ is always towards a fixed point $O$ on the line of motion of $P$. The acceleration is proportional to the displacement of $P$ from $O$.

**2** $\ddot{x} = v\dfrac{\mathrm{d}v}{\mathrm{d}x}$

**3** For a particle moving with **damped harmonic motion**

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + k\frac{\mathrm{d}x}{\mathrm{d}t} + \omega^2 x = 0$$

where $x$ is the displacement from a fixed point at time $t$, and $k$ and $\omega^2$ are positive constants.

**4** For a particle moving with **forced harmonic motion**

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + k\frac{\mathrm{d}x}{\mathrm{d}t} + \omega^2 x = \mathrm{f}(t)$$

where $x$ is the displacement from a fixed point at time $t$, and $k$ and $\omega^2$ are positive constants.

**5** You can solve coupled first-order linear differential equations by eliminating one of the dependent variables to form a second-order differential equation.